Group Coursework Submission Form

Specialist Masters Programme

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FACULTY OF ACTUARIAL SCIENCE AND INSURANCE

Analysis of the Swiss Market Index (SMI) from 2017 to 2023: Statistical Investigation of Logarithmic Returns and Financial Hypotheses

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1. Introduction

This study investigates the behavior of the **Swiss Market Index** (SSMI), using the daily closing prices over the period from **January 2017** to **December 2023**. This analysis's main goal is to understand the underlying behaviors and dynamics by focusing on the logarithmic returns. The motivation behind analysis is identifying patterns in the data, especially those that may be influenced by events such as the COVID-2019 pandemic, which introduced unusual market fluctuations and shutdowns.

By investigating these returns, we aim to conduct several statistical tests for volatility, distributional characteristics, and the normality of the returns. In addition, the analysis investigates the Markov Property, assessing whether past returns influence future returns. We examine the relation between the past and the future returns by testing the Efficient market hypothesis and potential predictability.

2. Data Cleaning and Normality Testing of SMI Logarithmic return

The analysis begins with the importing of the Swiss Market Index data from the csv file. The dataset contains two columns, namely, Date and Close (closing price). The statistics observed for the raw closing price data:

Min Value	1st Quartile	Median	Mean	3rd Quartile	Max Value
8161	9134	10301	10261	11123	12971

Table 1: *Raw Closing Price Data*

The graphical visualisation of the raw data is as below:

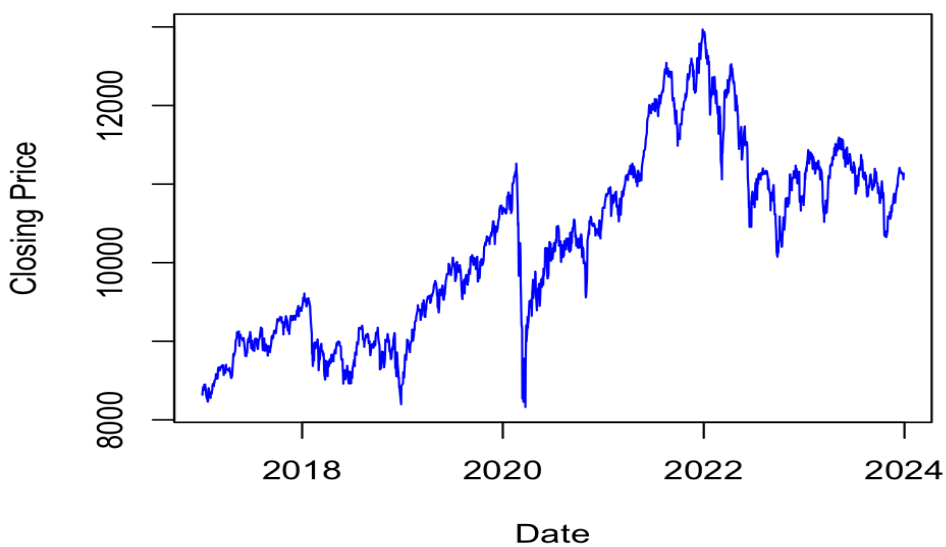

Fig. 1: *Raw SMI Data*

Logarithmic Transformation

To analyse the returns, we apply a logarithmic transformation to the closing prices. Then we also calculate the daily difference in the log prices, which is denoted by z_n.

The observed statistics for the logarithmic transformed data (z_n):

Min Value	1st Quartile	Median	Mean	3rd Quartile	Max Value
-0.1013393	-0.0043968	0.0005612	0.0001671	0.0048698	0.0678046

Table 2: *Logarithmic Closing Price Data*

Removal of Outliers

We further clean the logarithmic transformed (z_n) data for missing values, and we plot the graph, which shows there may be outliers.

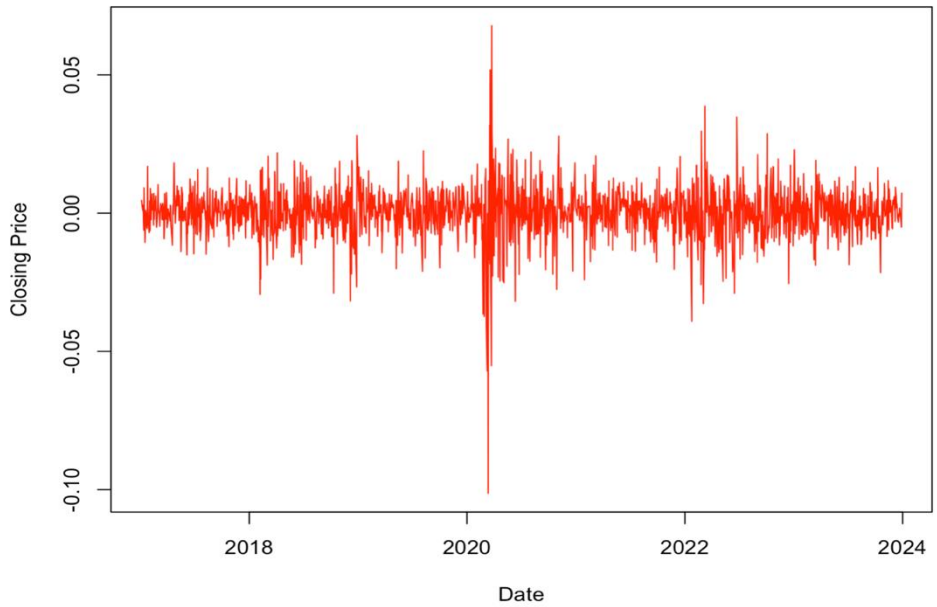

Fig. 2: *ln(Returns) plot*

We split the data into three segments (pre-, during and post- COVID), and attempt to remove outliers from the COVID era using the IQR method.

The following plots show the effects of our outlier removal.

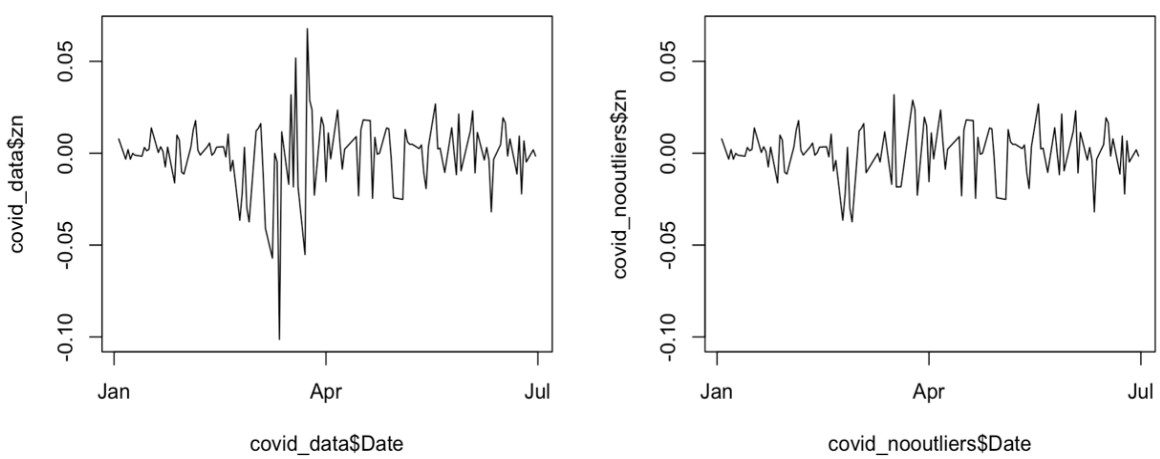

Fig. 3: *Covid Period ln(returns) pre- and post- Outlier Removal*

After recombining this outlier-free data with the pre- and post-COVID data sets, the data is much changed, as can be seen in the plots below:

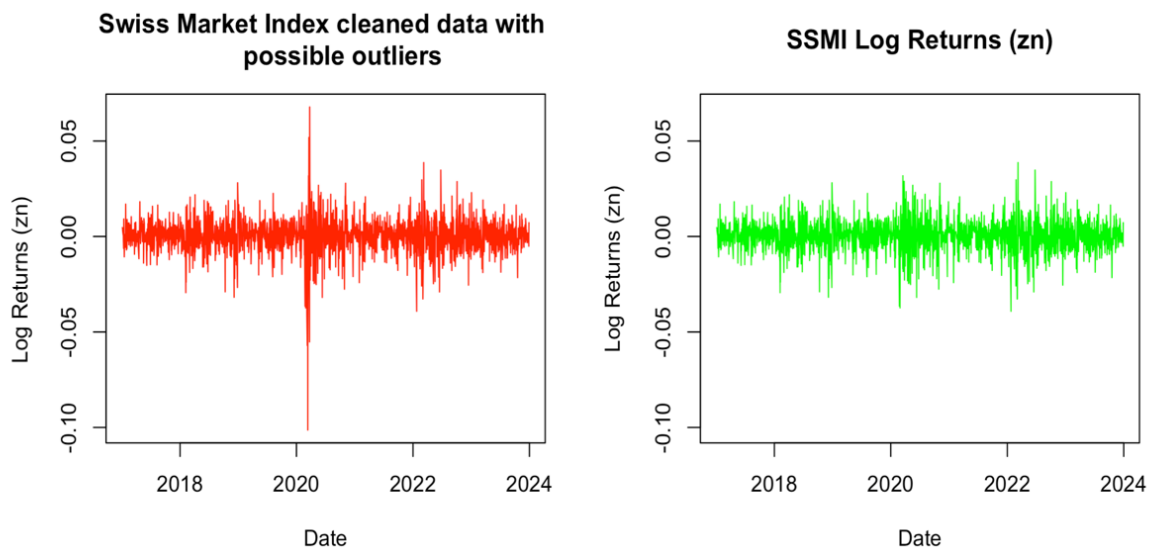

Fig. 4: *Yearly Variance Comparison of ln (Returns)*

The observed statistics for the cleaned outlier-free closing price data:

Mean	Variance	Min Value	Median	Max Value	Skewness	Excess Kurtosis
2.458×10^{-4}	7.118×10^{-5}	-0.0395	0.00056	0.0387	-0.234	-0.920

Table 3: *Outlier-free Logarithmic Closing Price Data*

Normality Tests

The Anderson-Darling and Kolmogorov-Smirnov normality tests were conducted on the data, and both test results showed *p*-values below 0.001, indicating the rejection of the null hypothesis that the log returns follow a normal distribution. This was further confirmed by the graphical analyses, including the Empirical Cumulative Distribution Function and the Q-Q plot, which showed significant deviations from the normal distribution characteristics.

The graphical visualisations using a Q-Q plot and an ECDF plot are shown as follows:

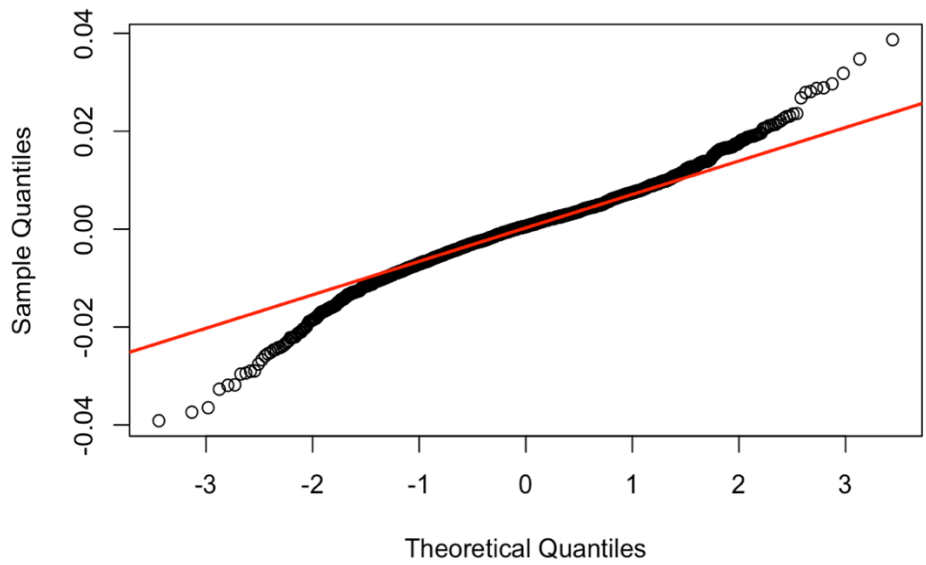

Fig. 5: *Q-Q Plot to Assess Normality*

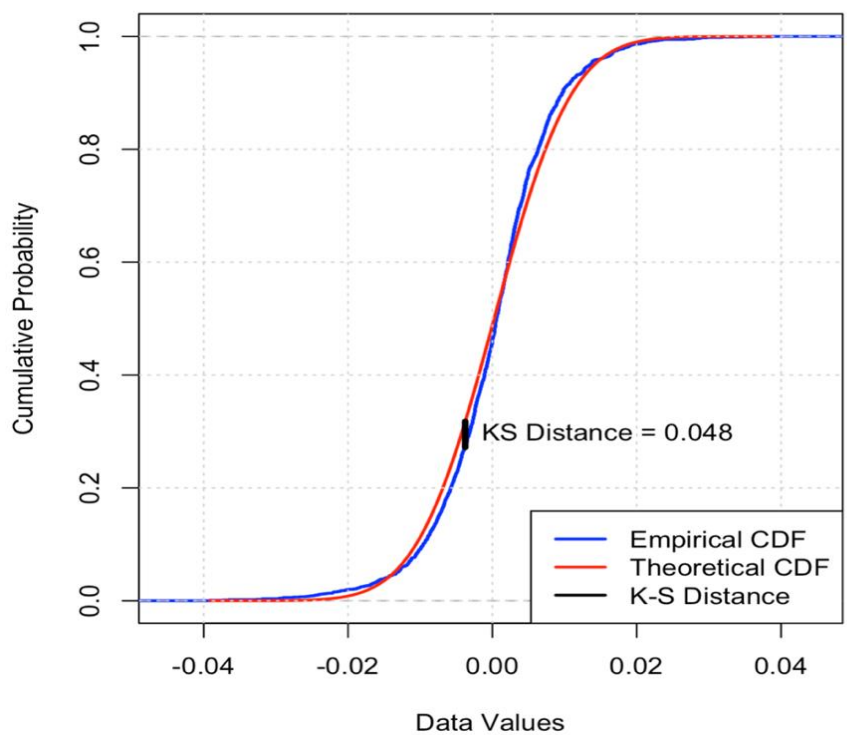

Fig. 6: *ECDF with Kolmogorov-Smirnoff Test*

These results suggest that traditional models assuming normality may not be suitable for SMI returns. Alternatively, we could use models that account for these characteristics, adjusting them to better understand market dynamics and make more informed financial decisions.

3. Investigating Normality by Resampling

Methodology

We calculate the excess kurtoses of 50,000 samples of size 12 from a normal distribution with mean and variance equal to the previously calculated sample mean and variance, and of 50,000 samples of size 12 taken directly from the *z*-data.

As we are using small samples, the calculation of their excess kurtoses using the method of moments estimator (as seen below) will produce biased estimates.

$$\hat{\gamma}_2 = \frac{m_4}{m_2^2} - 3 = \frac{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^4}{\left[\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2\right]^2} - 3$$

It is therefore better to "manually" calculate the excess kurtoses using the formula,

$$\hat{\gamma}_2 = \frac{n(n+1)}{(n-1)(n-2)(n-3)}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{s}\right)^4 - \frac{3(n-1)^2}{(n-2)(n-3)}$$ [1]

where *s* is the unbiased estimate of the sample standard deviation.

[1] Joanes, Derrick N.; Gill, Christine A. (1998), "Comparing measures of sample skewness and kurtosis", *Journal of the Royal Statistical Society, Series D*, **47** (1): 183–189

Results

Plotting the two sets of results on a histogram results in the following plot:

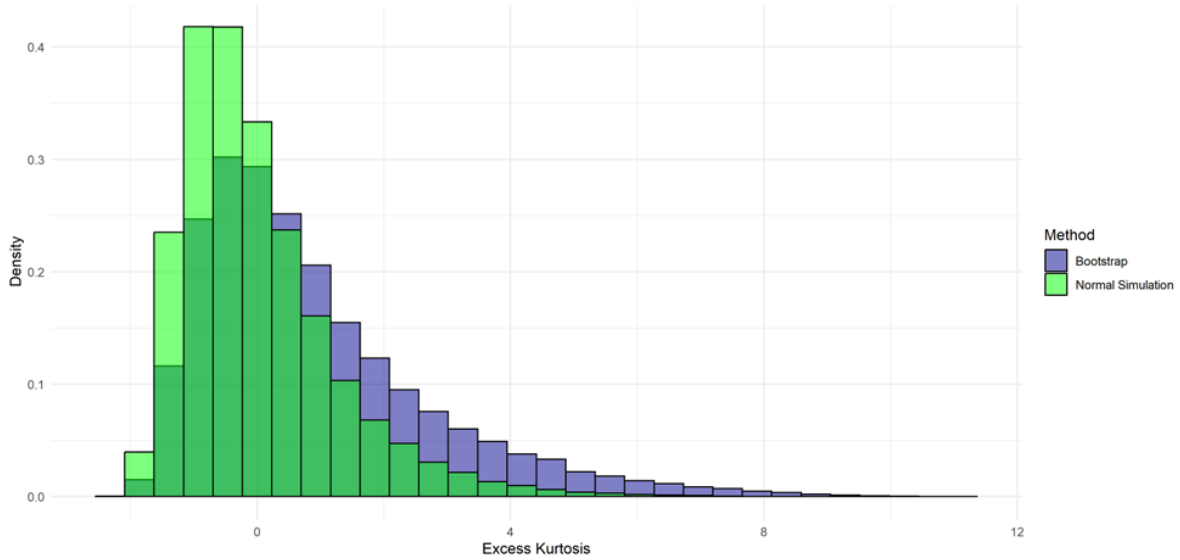

Fig. 7: *Bootstrap vs Normal Distribution Excess Kurtosis Histograms*

The histogram indicates that the bootstrapped sample had values that were consistently higher than the normal distribution's, suggesting that the *z*-data values may not be normally distributed, and that it has heavier tails. Both observations are in alignment with the results of the Anderson-Darling test (which indicated heavier tails) and Kolmogorov-Smirnov test.

4. Investigation of Constant Mean

After dividing the *z*-data into seven pieces, each representing one year, we plot box plots of each year to observe how the mean changes.

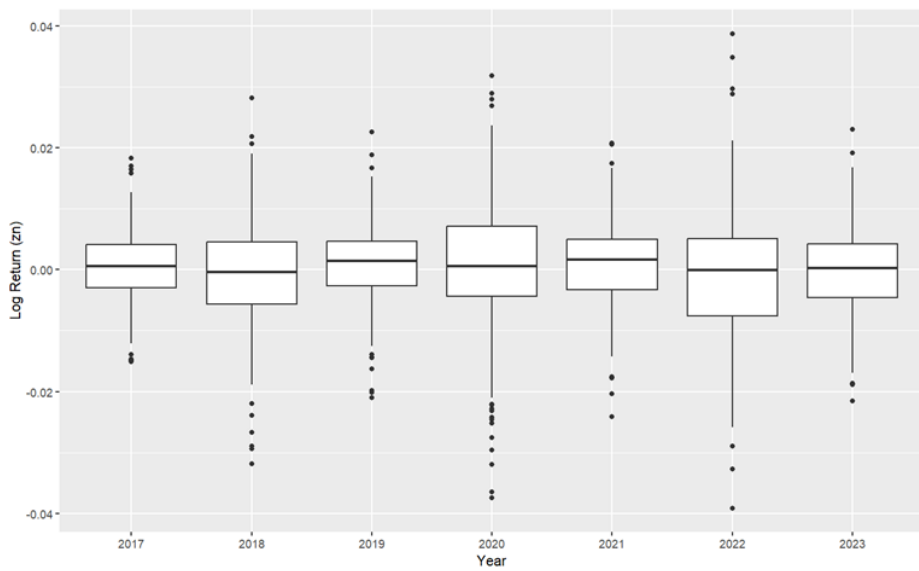

Fig. 8: *Boxplots of Yearly Means of ln(Returns)*

This diagram suggests that the mean logarithmic returns are similar year to year; an assumption of constant mean may be feasible.

If we then conduct the ANOVA test on the data, we see the following results:

Source	Degrees of Freedom	Sum of Squares	Mean Squares	F-Value	p-value
Year	6	0.00057	9.56×10^{-5}	1.345	0.234
Residuals	1735	0.12330	7.11×10^{-5}		

Table 4: *ANOVA Test Results*

The highlighted *p*-value suggests, at the 95% significance level, there is insufficient evidence to conclude that the means differ significantly over the years.

When conducting the Shapiro-Wilk test on the residuals, we obtain a *p*-value less than. This suggests they are not normally distributed. Although the assumption of normality is important for

ANOVA, as each of the years has on average around 250 observations, it is still reasonable to proceed with the ANOVA results (by the Central Limit Theorem).

Treating our data as a time series, the ADF (p-value < 0.01) and KPSS (p-value > 0.1) tests also suggest stationarity in the mean, supporting the ANOVA test's conclusion.

A plot of the rolling 30-day means of the data, with each year's mean value indicated by a red dashed line, affirms the assumption of constant mean.

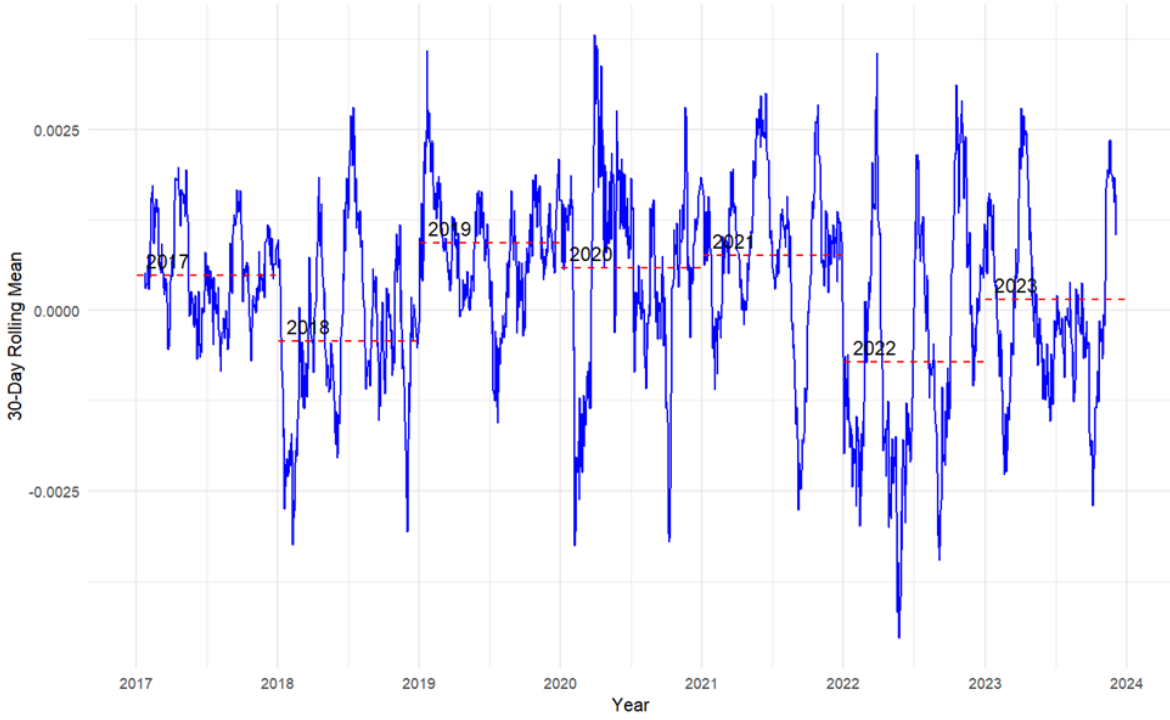

Fig. 9: *30-Day Rolling Mean of ln(Returns)*

5. Testing Variance Consistency Across Years

To assess whether the variance of the log returns (z_n) remained constant over the years, we began with a visual examination using boxplots. These illustrated the distribution of z_n for each year from 2017 to 2023. The variability in box heights suggested that the spread of z_n fluctuated across years.

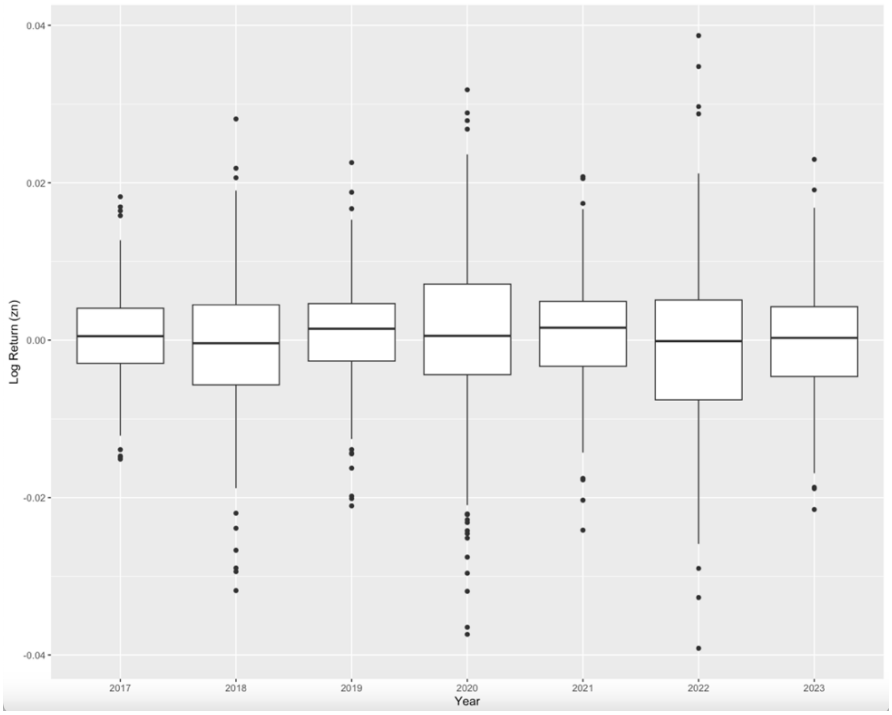

Fig. 10: *Yearly Variance Comparison of ln(Returns)*

Additionally, a 30-day rolling variance plot was generated to track changes in market volatility. The rolling variance highlighted notable periods of heightened volatility, such as during the COVID-19 pandemic in 2020–2021, as well as relatively stable phases.

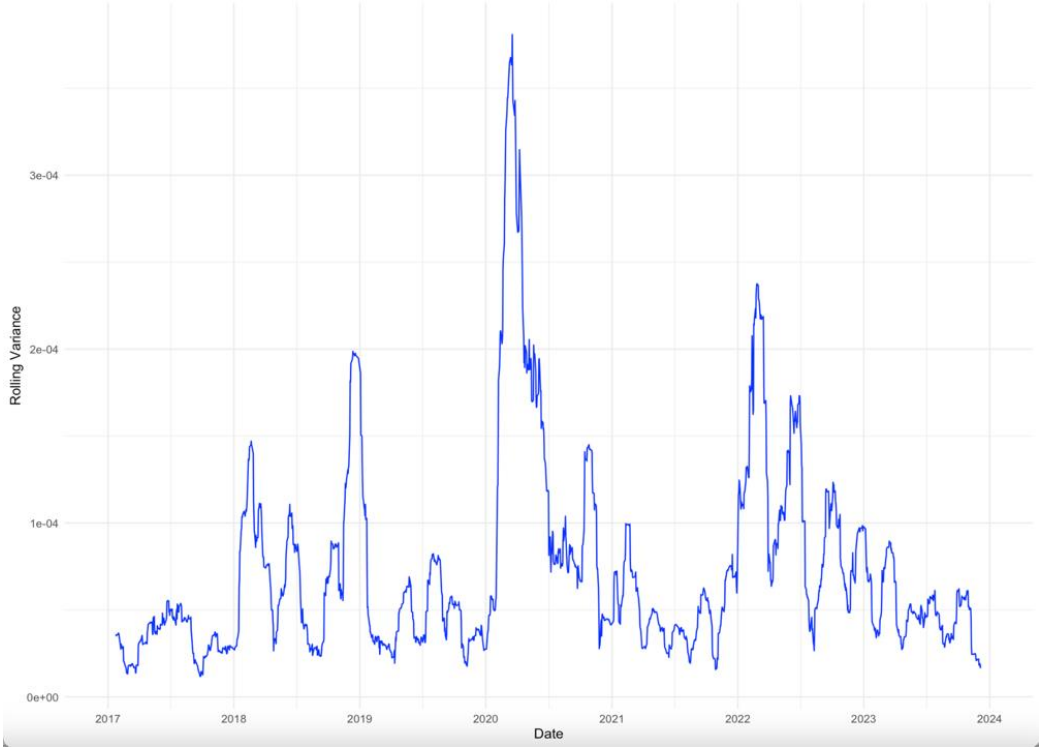

Fig. 11: *30-Day Rolling Variance of ln(Returns)*

Levene's Test was applied to the yearly subsets of z_n; it evaluates whether the variance is consistent across groups. The test results yielded a *p*-value below 0.05, providing strong evidence that the variances differed significantly across the years. The visual and statistical findings align with expectations that market volatility responds to external economic and geopolitical events.

6. Evaluating the Independence of Increments

To examine the independence of z_n from z_{n-1} , the log returns (z_n) were categorised into four groups based on quartiles. Each data point was assigned to a category (Q1, Q2, Q3, or Q4) depending on its magnitude. A similar categorisation was applied to z_{n-1}, creating a lagged variable $z_{n-1,\text{class}}$ for comparison.

A contingency table was constructed to display the frequencies of combinations between z_n and $z_{n-1,\text{class}}$ to see any dependencies between the current and previous day's log returns.

A chi-square test was performed on the contingency table to evaluate the statistical independence of z_n and z_{n-1}. The test results yielded a p-value greater than 0.05, indicating no significant association between the variables. This outcome supports the Efficient Markets Hypothesis (that daily returns are independent and driven by new information).

The findings reinforce the notion that market increments (z_n) are random and not influenced by previous increments (z_{n-1}).

7. The Markov Property

Call

lm(formula = z_n ~ z_n_lag1 z_n_lag2 + y_n_lag1 + y_n_lag2, data = cdata)

Residuals

Min Value	1st Quartile	Median	3rd Quartile	Max Value
-0.102013	-0.004547	0.000389	0.004784	0.065732

Table 5: *Residuals of Markov Property*

Coefficients

	Estimate	Std.error	t value	Pr(>|t|)
Intercept	0.036999	0.018118	2.042	0.0413
z_n_lag1	0.113814	0.368675	0.309	0.7576
z_n_lag2	-0.006493	0.024027	-0.270	0.7870
y_n_lag1	-0.134130	0.367608	-0.365	0.7153
y_n_lag2	0.130139	0.367627	0.354	0.7234

Table 6*: Coefficients of Markov Property*

- Residual standard error: 0.009258 on 1730 degrees of freedom
- Multiple R-squared: 0.002845 | Adjusted R-squared: 0.0005395
- F-statistic: 1.234 on 4 and 1730 DF, p-value: 0.2944

```
> cat("p-value for zn_lag2:", significant["zn_lag2"], "\n")
```

p-value for z_n_*lag2*: **0.7869998**

```
> cat("p-value for yn_lag2:", significant["yn_lag2"], "\n")
```

p-value for y_n_*lag2*: **0 .7233837**

Summary

We create a multiple regression model, with the first differences of log-transformed data (z_n) as the dependent variable and the lagged values of z_n and log-transformed data (y_n) as the independent variables.

Upon evaluation of the regression model, we see that the coefficients of the lagged variables z_n*_lag2* and z_n*_lag1* have *p*-values greater than 0.05, indicating that they don't contribute to the forecasting of z_n.

If the Markov property holds, merely z_n*_lag1* and y_n*_lag1* should be enough for the best z_n forecast. However, the high dependence on z_n*_lag2* and y_n*_lag2* indicates that the data may not fully satisfy the Markov property.

8. Conclusion

Our analysis of the Swiss Market Index (SMI) log returns from 2017 to 2023 explored key revealed non-normality, significant variance inconsistencies over time, and independence of daily increments, supporting the Efficient Markets Hypothesis.

The Markov property was evaluated through regression, showing lagged variables beyond one day were insignificant ($p > 0.05$), indicating no additional predictive power. These results suggest that SMI returns exhibit random, unpredictable behavior, aligning with efficient market theory. The findings provide valuable insights into financial dynamics.

9. Use of Generative AI Tools

Generative AI tools (like ChatGPT, Julius AI and Gemini) were used for a variety of different tasks in this project. The specific uses include data interpretation (e.g. for understanding the purpose of the Shapiro-Wilk test, and how to interpret their results), suggestions for code (certain graphs were modified using suggestions obtained from AI), grammar checks and formatting advice.